

Predicting the magnetic moments of materials from material composition.

Magnetic materials are widely used in various areas such as data storage, medical imaging, electric motors. This project investigates whether the magnetic moments of materials can be predicted from simple compositional features through machine learning without needing Density Functional Theory (DFT) which has calculation that demand significant computational resources.

The main dataset will be gathered from Materials Project database using the mp-api Python client. The dataset will include approximately 20,000 inorganic compounds sampled across four magnetic ordering types: ferromagnetic (FM), antiferromagnetic (AFM), ferrimagnetic (FiM), and non-magnetic (NM) ~ 5,000 compounds each to make sure that the dataset is balanced and diverse enough. Structural features collected will consist of chemical formula, number of sites, density, bandgap, total magnetization normalized by formula units, magnetic ordering type.

For data enrichment pymatgen and matminer libraries will be used. Structural features will be parsed and computed using pymatgen. Different material properties such as electronegativity, magnetic element fraction, valence electron count etc. will be calculated using these libraries. Compositional features (features calculated via formulas) will be calculated using matminer's ElementProperty featurizer which automatically generates statistics of different elemental properties such as electronegativity, d-orbital electron count, valence electron count. Additional derived features, such as magnetic element fraction, may also be included. These features will capture the main physical and chemical principles that drive magnetic behavior in materials.

The goal is to capture the physical and chemical factors that influence magnetic behavior using only composition-based information. The final dataset is expected to contain around 20,000 samples. The primary target variable of the study is total magnetization normalized by formula units (μB per formula unit).